Cost Optimization, Risk Analysis, and Dynamic Stopping for Random Blind-Draw Collections

A Case Study of the *DEATH NOTE – My Dear Light*
PARCO Pop-Up Badge Series

Mathematical Modeling Project Report

October 2026

Contents

Abstract **III**

1 Introduction **1**

1.1 Background . . . 1
1.2 Objectives and Significance . . . 1
1.3 Research Questions . . . 2
1.4 Related Models and Positioning . . . 2
1.5 Modeling Roadmap . . . 2

2 Data Collection and Analysis **3**

2.1 Case Study and Data Sources . . . 3
2.2 Sampling Protocol and Record Audit . . . 3
2.3 Price Characteristics . . . 4

3 Model Assumptions and Notation **5**

3.1 Assumptions . . . 5
3.2 Notation . . . 6

4 Model Development: From Static Stopping to Dynamic Decisions **6**

4.1 Unified Cost Convention . . . 6
4.2 Boundary Baselines . . . 7
4.3 Fixed-Stop Model . . . 7
4.4 State-Dependent One-Step Look-Ahead Rule . . . 8
4.5 Bellman Optimal-Stopping Model . . . 9
4.6 Solution Methods and Consistency Checks . . . 10

5 Model Solution and Results **10**

5.1 Baseline Parameters . . . 10
5.2 Draw-Until-Complete Baseline . . . 11
5.3 Fixed-Stop Strategy . . . 12
5.4 Price-Aware Dynamic Stopping . . . 14
5.5 Unified Expected-Cost Comparison . . . 16
5.6 Analytical and Numerical Bellman Verification . . . 16
5.7 Monte Carlo Risk Analysis . . . 19
5.8 Agreement Between Theory and Simulation . . . 21
5.9 Risk-Based Recommendations . . . 21

6 Sensitivity Analysis and Model Discussion **22**
6.1 Effect of Resale Success Probability . . . 22
6.2 Effect of the Secondary-Market Price Level . . . 22
6.3 Effect of Blind-Draw Cost . . . 22
6.4 Transaction Frictions . . . 23
6.5 Joint Sensitivity of Draw Price and Resale Success . . . 24
6.6 Unequal Draw Probabilities . . . 25
6.7 Price-Sample Uncertainty . . . 26
6.8 Conservative Check of Flagged Price Records . . . 26
6.9 Shared Marginal Principle of Fixed and Dynamic Rules . . . 27
6.10 Sampling-Mechanism Boundary . . . 27
6.11 Model Strengths . . . 28
6.12 Model Limitations . . . 28

7 Conclusion **29**

References **30**

A Monte Carlo Simulation Procedure **31**
A.1 Fixed-Stop Strategy . . . 31
A.2 Price-Aware Dynamic Strategy . . . 31

B Supplementary Diagnostic Figures **32**

C All Two-Design Dynamic States **34**

D Key Results and Reproducibility **35**

E Price-Record Traceability **36**

Abstract

Collectors seeking a complete set must choose between buying more random packs and purchasing the missing designs directly. Duplicates and price differences across designs affect both cost and risk. Using the eight-design *DEATH NOTE – My Dear Light* PARCO badge series, we compare a fixed draw limit with decisions based on which designs have been collected.

We reviewed 134 secondary-market listing records dated mainly in August 2026. After excluding seven bundles without a reliable unit price and two records with missing prices, 125 observations remained. We estimated purchase costs using each design's median listing price. All strategies share the same net-cost definition: spending on draws plus purchases of missing designs, minus resale income from duplicates. Duplicate resale succeeds independently with a specified probability.

Under independent, equal-probability draws, the coupon collector model provides a draw-until-complete benchmark for comparison with direct purchase. A fixed-stop strategy buys any missing designs after exactly k draws; we derive its expected cost. With a draw price of CNY 39.8 and a resale success probability of 0.60, direct purchase costs CNY 366.58. Drawing until completion requires 21.74 draws on average and has an expected net cost of CNY 571.95. Fixed $k = 3$ minimizes expected net cost at CNY 357.31; the expectation for $k = 2$ is only CNY 0.29 higher.

The dynamic rule uses the set of missing designs to compare the expected value of another draw with its price. A Bellman analysis confirms optimality for expected net cost under fixed prices, probabilities, and resale terms. The proof requires expected resale recovery not to exceed purchase cost for each design, and the draw price to exceed expected recovery when every draw is a duplicate. At the baseline inputs, drawing continues until two distinct designs have appeared. Among the 28 possible pairs of acquired designs, the policy continues for 10 and stops for 18; it also stops when a third distinct design appears. The rule and Bellman policy agree in all 256 states. From an empty collection, the expected draw count is 2.619 and expected net cost is CNY 356.72.

We ran 100,000 Monte Carlo trials per randomized strategy to compare variability and tail risk. The dynamic strategy's simulated mean was CNY 356.81. Estimated probabilities that final net cost exceeds CNY 400 are 3.49%, 7.27%, and 6.20% for fixed $k = 2$, fixed $k = 3$, and dynamic stopping, respectively. In 2,000 within-design bootstrap resamples, $k = 2$ minimizes expected cost in 53.35% and $k = 3$ in 45.15%; decisions for 21 of the 28 acquired-design pairs are sensitive to the resampled prices. The small mean-cost advantage therefore does not guarantee lower tail risk or a stable stopping decision. These results apply to independent loose packs; purchasing a verified nonrepeating box follows a different cost model.

Keywords: random blind draw; Coupon Collector problem; fixed stopping; dynamic stopping; Bellman equation; Monte Carlo simulation; risk analysis

1 Introduction

1.1 Background

"Guzi" usually refers to merchandise such as badges, acrylic stands, and cards based on characters from anime, games, movies, and TV shows. Some products are sold in random packages, so consumers do not know which character they will receive before making a purchase. For collectors who want to complete a full set, random purchases may create duplicate badges and missing badges. At the same time, different characters may have very different prices in the second-hand market. Therefore, the cost of completing a collection is not determined only by the official selling price.

Some official boxed products may use a configuration in which no duplicate items appear within one box. However, consumers cannot always obtain a complete box directly. In situations such as pop-up store sales, reseller box splitting, and single-pack resale, consumers often buy random packs whose origins cannot be fully tracked. Therefore, this study focuses on a fully mixed single-pack market and models each purchase as an independent random blind purchase.

Collectors must decide whether to continue blind draws or buy the missing designs directly on the secondary market. Another draw may obtain an expensive missing design, but it may also produce a duplicate. We compare these choices using expected net cost and report their tail risks separately.

1.2 Objectives and Significance

We study how collection state, completion prices, and duplicate recovery affect the stopping decision. The main objectives are:

(1) characterize the expected net cost and right-tail risk of drawing until every design has appeared;

(2) derive a fixed-stop model and identify the prespecified draw count k that minimizes expected net cost;

(3) incorporate uncertain resale of duplicates under the same accounting convention in theory and simulation;

(4) quantify standard deviation, P90, P95, and budget-exceedance probability through Monte Carlo simulation;

(5) extend the fixed rule to a state-dependent, price-aware one-step look-ahead policy;

(6) formulate a Bellman optimal-stopping model and verify the simpler rule both analytically and by exhaustive state enumeration; and

(7) test whether the recommendations remain stable when prices, draw cost, resale success, and transaction frictions change.

A fixed draw limit does not distinguish between collections containing different designs. Because secondary-market prices vary across designs, we extend the model to make the stopping decision depend on the observed collection.

1.3 Research Questions

This study focuses on the following five research questions:

RQ1. How should secondary-market observations be cleaned to estimate representative purchase costs?

RQ2. How does a prespecified draw count k affect expected total cost, and why does an optimal k^* arise?

RQ3. How do secondary-market-only, fixed-stop, and draw-until-complete strategies differ in expected cost and risk?

RQ4. Can the observed set of acquired designs support a more precise dynamic stopping decision?

RQ5. How stable are the strategy boundaries when prices, draw cost, resale success, and transaction frictions change?

1.4 Related Models and Positioning

The classical Coupon Collector problem characterizes the waiting time required to observe every category under repeated sampling; equal- and unequal-probability variants are well established [1–5]. Optimal stopping and dynamic programming provide a general framework for choosing between continuation and termination after observing the current state [6, 7]. This study combines these ideas with heterogeneous completion prices and uncertain resale recovery.

1.5 Modeling Roadmap

All strategies use the same net-cost convention: blind-draw spending plus purchases of missing designs, minus realized resale income. Direct purchase and drawing until completion provide the two baselines. The fixed-stop model compares predetermined draw limits, while the dynamic model makes the stopping decision from the observed missing-design set. A Bellman recursion checks whether the one-step rule accounts for future decisions. Simulation compares cost variability and tail risk, and sensitivity analysis tests how the recommendations change with the inputs. Figure 1 summarizes the models and their checks.

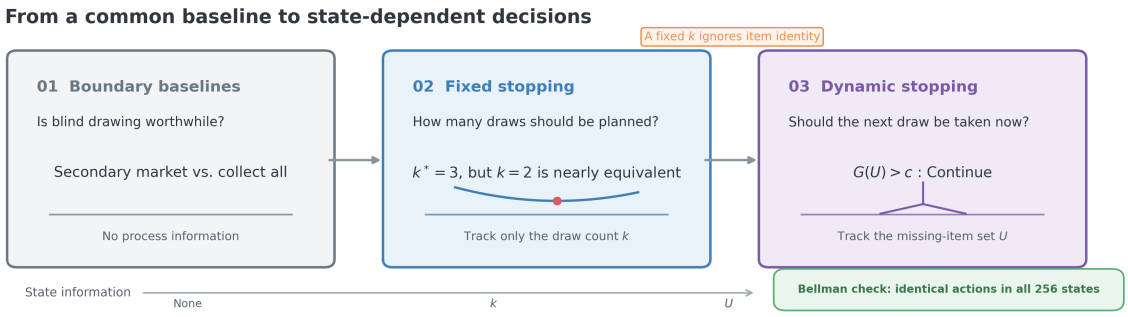


Figure 1. Progression from boundary baselines and fixed stopping to dynamic stopping and Bellman verification.

2 Data Collection and Analysis

2.1 Case Study and Data Sources

The case is the *DEATH NOTE – My Dear Light* matte badge series sold at a PARCO pop-up event. The official press release lists eight designs, a single-package price of JPY 550 including tax, and a full-box price of JPY 4,400 [8]. The historical PARCO event-detail link is retained for source traceability [9]. After allowing for purchasing channel, exchange rate, and cross-border shipping, we use CNY 39.8 as a scenario value for one blind draw in China. It is not estimated from the secondary-market sample.

The secondary-market data consist of Xianyu listing records dated mainly from August 5 to August 19, 2026 [10]. The original workbook records source URLs, evidence identifiers, and intended screenshot filenames. These fields permit a record-level audit, but a filename alone does not verify an archived screenshot or a completed transaction. The eight designs depict two versions each of Light Yagami, L, and Misa Amane, plus Near and Mello.

2.2 Sampling Protocol and Record Audit

The workbook is a convenience sample of available listings, not a random sample of market transactions. Its eligibility fields distinguish the target series and specification, item condition, bundle composition, abnormal-price flags, and inclusion status. The calculation file preserves the workbook's buyer-paid unit costs and uses only records marked for inclusion with a numeric unit price. We do not impose an additional numerical outlier cutoff: high or low prices that pass these recorded checks remain in the sample.

We screened 134 listing records. Seven heterogeneous bundles and two observations with missing price fields were excluded, leaving 125 usable records. Since the distributions were skewed and contained a small number of extreme values, the within-design median was used as b_i. This is a simple application of robust location estimation for small market samples [11].

The 125 usable records have 125 distinct evidence identifiers but only 114 distinct source URLs. Several listings cover more than one design, and records MKT-068 and MKT-075 share both a design and a URL. Seller identifiers and search keywords were not systematically recorded, so complete seller-level deduplication and the original search sequence cannot be reconstructed. We retain the original records for the baseline comparison and separately examine a conservative exclusion check in Section 6.8.

One record, MKT-126, is dated August 19, 2025, whereas the remaining usable records are dated August 5–19, 2026. We retain the recorded year because we cannot verify whether it is correct. Appendix E gives examples of our evidence mapping and describes the data limitations. Our mapping file links each usable record to its row in the English workbook, source URL, recorded date, and screenshot filename. These links make the price extraction reproducible, but do not verify the sampling process or the source pages.

2.3 Price Characteristics

Table 1. Secondary-market price statistics for the eight badge designs; monetary values are in CNY.

Design	Records	Median b_i	Mean	Range
Light Yagami, front view	16	60.00	57.19	44.00–70.00
Light Yagami, profile	17	58.00	61.18	45.00–80.00
L, light-gray portrait	20	60.00	56.49	37.00–76.00
L, dark-gray profile	14	58.50	57.06	45.00–75.00
Misa Amane, smiling	14	37.75	37.95	27.50–52.00
Misa Amane, choker	14	32.00	32.64	20.00–60.00
Near	17	20.33	23.11	14.08–33.00
Mello	13	40.00	41.54	28.00–68.00

The representative purchase costs sum to

$$B = \sum_{i=1}^{8} b_i = 366.58 \text{ CNY}.$$

Figure 2 shows the raw observations, medians, bootstrap intervals, and expected resale values. The cross-design price differences provide information that the dynamic model can use.

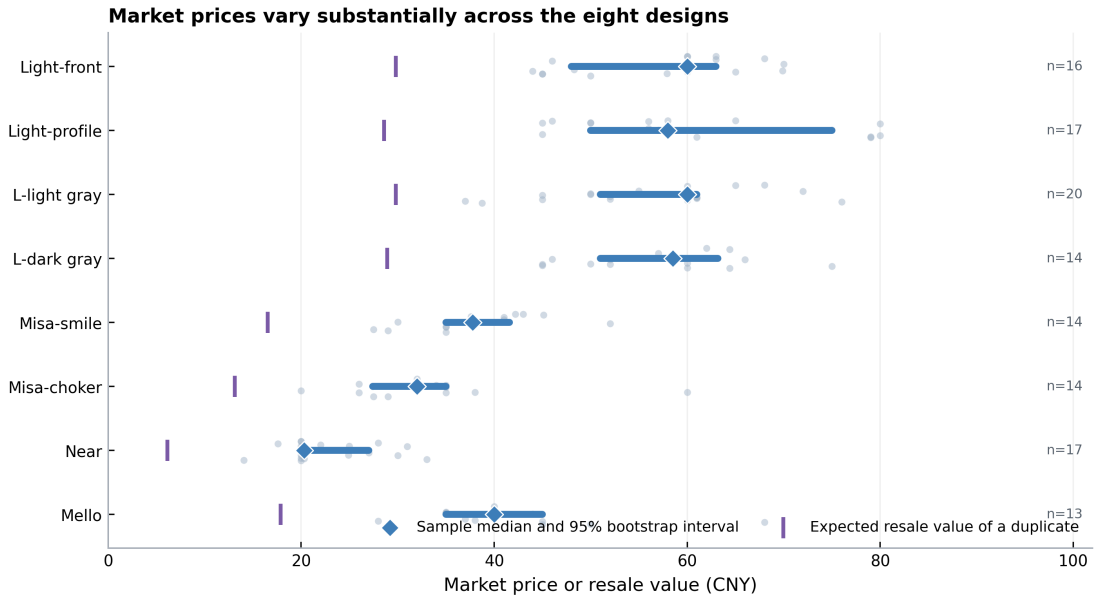


Figure 2. Raw price observations, sample medians, 95% bootstrap intervals, and expected resale values of duplicates.

3 Model Assumptions and Notation

3.1 Assumptions

Assumption 1: Independent, equal-probability blind draws. We approximate purchases from mixed loose packs as independent draws, with each of the eight designs occurring with probability $1/8$. These probabilities are modeling assumptions rather than estimates from observed draw frequencies. Purchases from a verified nonrepeating box are considered separately.

Assumption 2: Short-run market stability and availability. All designs are assumed to remain available during the decision period at their representative purchase costs b_i. We estimate these costs by the within-design medians. Temporary shortages, waiting costs, and price changes within a decision period are excluded.

Assumption 3: Independent resale of duplicates. A duplicate of design i sells with probability η_i. A successful sale yields net revenue g_i; otherwise, recovery is zero within the study period. Resale outcomes are independent across duplicates.

Assumption 4: Fully observed, stationary state. After each blind draw, the collector can identify the obtained badge design and update the current set of missing badges. Dynamic decisions are based only on the current collection status. The strategy does not attempt to predict future market prices. Draw probabilities, second-hand prices, and resale rules are assumed to remain unchanged during the decision period.

Assumption 5: Monetary evaluation. This study evaluates strategies only by monetary cost and risk measures. The comparison focuses on total net cost, cost variability, and the probability of budget overruns. Non-monetary factors such as opening experience, emotional value, time cost, social exchange value, and character preference are not included in the model.

3.2 Notation

Table 2 summarizes the principal symbols used in the models.

Table 2. Main notation.

Symbol	Definition
n	Number of target designs; $n = 8$ in the case study.
p_i	Probability of drawing design i.
c	Cost of one blind draw.
b_i	Representative secondary-market purchase cost of design i.
B	Cost of purchasing the full set, $B = \sum_i b_i$.
k	Prespecified number of blind draws in a fixed-stop strategy.
T	Number of draws required to obtain the complete set.
$N_i(k)$	Number of copies of design i obtained in k draws.
$I_i(k)$	Indicator that design i is missing after k draws.
$D_i(k)$	Number of duplicates of design i after k draws.
d	Shipping component removed from buyer-paid price when estimating resale revenue.
γ	Platform fee rate.
h	Seller-side packaging and shipping cost.
g_i	Net revenue if one duplicate of design i is successfully sold.
η_i	Probability that a duplicate of design i is sold.
r_i	Expected net recovery from one duplicate, $r_i = \eta_i g_i$.
R	Sum of expected recoveries, $R = \sum_i r_i$.
U	Current missing-item set.

4 Model Development: From Static Stopping to Dynamic Decisions

We first define the cost components shared by all strategies, then derive the collect-all and fixed-stop models. The dynamic model uses the identities of missing designs to choose between another draw and secondary-market completion.

4.1 Unified Cost Convention

Let p_i be the probability of drawing design i, with $\sum_i p_i = 1$, and let c be the paid cost of one blind draw. If the collector stops while design i is still missing, that design is purchased for b_i; hence a complete secondary-market purchase costs $B = \sum_i b_i$.

The resale side must use the same price convention as the purchase side. Because b_i is a buyer-paid total, the amount retained by a seller is not simply b_i. If a duplicate of design i is sold, its seller-side net revenue is approximated by

$$g_i = \max\{(1-\gamma)(b_i - d) - h, 0\}, \qquad r_i = \eta_i g_i. \tag{4.1}$$

Every model evaluates

$$\text{net cost} = \text{blind-draw spending} + \text{completion purchases} - \text{realized resale income}.$$

Theoretical expectations use r_i directly. Simulations generate a Bernoulli(η_i) resale event for each duplicate, maintaining the same economic convention.

We also record purchase outlay before subtracting resale income. If the stopping time is τ and the missing set at stopping is U_τ, define

$$A = \tau c + B(U_\tau), \qquad C = A - Y, \tag{4.2}$$

where Y is realized net recovery from duplicates. Under the scenario in which resale proceeds arrive only after blind draws and completion purchases have been paid for, A is the amount spent on those purchases before recovery. It excludes separately timed resale expenses and does not model a full cash-flow schedule. The probabilities $\mathbb{P}(C > L)$ and $\mathbb{P}(A > L)$ therefore answer different questions. Neither is optimized as a hard-budget constraint by the expected-cost model.

4.2 Boundary Baselines

4.2.1 Complete Secondary-Market Purchase

Buying all designs without blind draws produces the deterministic cost

$$C_S = B = \sum_{i=1}^{n} b_i. \tag{4.3}$$

4.2.2 Draw Until Complete

When m distinct designs have been collected, the probability that the next draw is new is

$$q_m = \frac{n-m}{n}.$$

If Y_m is the waiting time to move from m to $m+1$ distinct designs, then Y_m is geometric with mean $1/q_m$ [12]. Hence

$$T = \sum_{m=0}^{n-1} Y_m, \qquad \mathbb{E}[T] = n \sum_{j=1}^{n} \frac{1}{j} = nH_n. \tag{4.4}$$

By label symmetry, the expected number of duplicates of each design at completion is $H_n - 1$. Therefore

$$\mathbb{E}[C_{AR}] = cnH_n - R(H_n - 1). \tag{4.5}$$

4.3 Fixed-Stop Model

A fixed-stop strategy chooses an integer k before any purchases. After exactly k blind draws, every missing design is bought on the secondary market. Let $N_i(k)$ count the copies of design i obtained in those draws.

Define

$$I_i(k) = \mathbf{1}_{\{N_i(k)=0\}}, \qquad D_i(k) = \max\{N_i(k) - 1, 0\}.$$

For one realization,

$$C_{FR}(k) = kc + \sum_{i=1}^{n} b_i I_i(k) - \sum_{i=1}^{n} \sum_{\ell=1}^{D_i(k)} Z_{i\ell} g_i, \tag{4.6}$$

where $Z_{i\ell} \sim$ Bernoulli(η_i). Since $N_i(k) \sim$ Binomial(k, p_i),

$$\mathbb{P}(I_i(k) = 1) = (1 - p_i)^k, \qquad \mathbb{E}[D_i(k)] = kp_i - 1 + (1 - p_i)^k.$$

Thus

$$\mathbb{E}[C_{FR}(k)] = kc + \sum_{i=1}^{n} b_i (1 - p_i)^k - \sum_{i=1}^{n} r_i \left[kp_i - 1 + (1 - p_i)^k \right]. \tag{4.7}$$

Under equal probabilities, let $q = (n - 1)/n$. The expression reduces to

$$\mathbb{E}[C_{FR}(k)] = kc + Bq^k - R\left(\frac{k}{n} - 1 + q^k\right), \tag{4.8}$$

and the adjacent difference is

$$\Delta_k = \mathbb{E}[C_{FR}(k + 1)] - \mathbb{E}[C_{FR}(k)] = c - \frac{R}{n} - \frac{B - R}{n} q^k. \tag{4.9}$$

When $B > R$ and $R/n < c$, Δ_k increases monotonically and eventually becomes positive. The first nonnegative adjacent difference identifies the integer optimum.

4.4 State-Dependent One-Step Look-Ahead Rule

A fixed draw limit ignores which designs have appeared. After two draws, a collector who owns two expensive designs faces a different completion cost from one who owns two inexpensive designs. We therefore base the dynamic decision on the missing-design set.

Let $\mathcal{I} = \{1, \ldots, n\}$ and

$$U_t = \{i : N_i(t) = 0\}, \qquad O_t = \{1, \ldots, n\} \setminus U_t.$$

If the process stops in state U, the completion cost is

$$B(U) = \sum_{i \in U} b_i. \tag{4.10}$$

Under the risk-neutral expected-cost objective, U_t is a sufficient Markov state. Past draw spending is sunk and does not affect the comparison between future actions. With stationary probabilities and additive resale recovery, the future cost distribution depends on which designs are missing, not on the order in which existing designs were obtained. The exact number of past duplicates also need not be stored because every additional duplicate of design i contributes the same expected recovery r_i. This sufficiency would fail under a hard remaining budget, time-varying prices, inventory limits, or a nonlinear risk utility; such extensions would require additional state variables.

If the next draw yields a missing design $i \in U$, it avoids a future purchase of b_i. If it yields an already-owned design $i \notin U$, it generates expected resale recovery r_i. The expected economic value of one additional draw is therefore

$$G(U) = \sum_{i \in U} p_i b_i + \sum_{i \notin U} p_i r_i. \tag{4.11}$$

Comparing this value with c gives the interpretable candidate policy

$$\pi_G(U) = \begin{cases} \text{Continue}, & G(U) > c, \\ \text{Stop}, & G(U) \leq c. \end{cases} \tag{4.12}$$

Equality is assigned to Stop because an additional draw adds variability without lowering expected cost.

The random stopping time induced by this rule is

$$\tau_G = \inf\{t \geq 0 : G(U_t) \leq c\}.$$

Comparing $G(U)$ with c gives a candidate stopping rule based on the next draw. This comparison alone does not establish whether later draws could lower expected cost. We use the Bellman model to account for those future decisions.

4.4.1 Equal-Probability State Threshold

For equal probabilities, define $\delta_i = b_i - r_i$ and $O = \mathcal{I} \setminus U$. Then

$$G(U) = \frac{1}{n}\left(B - \sum_{i \in O} \delta_i\right),$$

so the rule becomes

$$\sum_{i \in O} \delta_i < B - nc \Rightarrow \text{Continue}, \qquad \sum_{i \in O} \delta_i \geq B - nc \Rightarrow \text{Stop}. \tag{4.13}$$

Under the baseline parameters, $B - nc = 48.18$.

Acquiring design i reduces the gain $G(U)$ in Eq. (4.11) by δ_i/n, since a later draw of that design yields expected resale recovery rather than avoiding a missing-item purchase. Once $\sum_{i \in O} \delta_i$ reaches 48.18, the expected gain from another draw no longer covers its price.

4.5 Bellman Optimal-Stopping Model

The one-step rule is interpretable, but global optimality must account for future decisions. Let $V^*(U)$ be the minimum expected remaining net cost from state U, with $V^*(\varnothing) = 0$. The implicit Bellman equation is

$$V^*(U) = \min\left\{ B(U),\ c + \sum_{i \in U} p_i V^*(U \setminus \{i\}) + \sum_{j \notin U} p_j \big(V^*(U) - r_j\big) \right\}. \tag{4.14}$$

Let $q(U) = \sum_{i \in U} p_i$. If continuation is optimal, the self-loop in Eq. (4.14) can be removed:

$$C^*_{\text{cont}}(U) = \frac{c + \sum_{i \in U} p_i V^*(U \setminus \{i\}) - \sum_{j \notin U} p_j r_j}{q(U)}. \tag{4.15}$$

Therefore the explicit recursion is

$$V^*(U) = \min\{B(U), C^*_{\text{cont}}(U)\}. \tag{4.16}$$

Because every successor state in the numerator has one fewer missing design, the values can be computed by backward induction over $|U|$.

The self-loop elimination also clarifies the action comparison. If one substitutes the stopping value $V^*(U) = B(U)$ into the implicit continuation branch, the difference from stopping is $q(U)[C^*_{\text{cont}}(U) - B(U)]$. Since $q(U) > 0$ for every nonterminal state, the implicit and explicit formulations select exactly the same action.

To evaluate the interpretable rule independently, define its policy value $V_{\pi_G}(U)$. When $\pi_G(U)$ stops, $V_{\pi_G}(U) = B(U)$; when it continues,

$$V_{\pi_G}(U) = \frac{c + \sum_{i \in U} p_i V_{\pi_G}(U \setminus \{i\}) - \sum_{j \notin U} p_j r_j}{q(U)}. \tag{4.17}$$

Thus the policy value V_{π_G} in Eq. (4.17) answers how much the interpretable rule costs, whereas V^* is the minimum attainable value over all state-dependent actions. Agreement between their actions in every state establishes optimality of the simpler rule for the current model.

4.6 Solution Methods and Consistency Checks

We use analytical formulas, state recursion, and simulation for different parts of the solution. First, closed-form expectations and adjacent differences solve the boundary and fixed-stop models. Second, memoized recursion evaluates all $2^8 = 256$ missing-item sets and compares the one-step and Bellman actions state by state. Third, Monte Carlo simulation explicitly generates draws and resale outcomes to estimate dispersion, quantiles, and budget risk [13, 14]. A within-design price bootstrap then propagates sampling uncertainty through both the fixed and dynamic optimizations.

5 Model Solution and Results

5.1 Baseline Parameters

We estimate each purchase cost b_i by the within-design median. The remaining inputs define the baseline decision environment and are examined through sensitivity analysis. Table 3 distinguishes these scenario inputs from the observed prices.

Table 3. Source and uncertainty treatment of baseline parameters.

Parameter	Source or status	Uncertainty treatment
b_i	Within-design medians from 125 usable listings	Within-design nonparametric bootstrap
$c = 39.8$	Scenario value based on the JPY 550 official price and a China purchase convention	Scanned from CNY 35 to 45
$\eta = 0.60$	Baseline assumption without transaction-level time-to-sale data	Scanned from 0.35 to 0.80
$\gamma = 0.006$	Baseline fee-rate scenario; no archived rule is supplied	Scanned from 0 to 0.05
$d = 6,\ h = 4$	Scenario values for shipping, packaging, and seller effort	Local one-way perturbations
CNY 400	Integer reference slightly above $B = 366.58$	Full budget–exceedance curve also reported

Table 4 lists purchase costs and the recovery values computed using Eq. (4.1). The numerical baseline is

$$n = 8, \qquad c = 39.8, \qquad \eta_i = 0.60, \qquad \gamma = 0.006, \qquad d = 6, \qquad h = 4.$$

Table 4. Purchase costs and duplicate-recovery parameters (CNY).

Design	b_i	g_i	r_i
Light Yagami, front view	60.00	49.676	29.806
Light Yagami, profile	58.00	47.688	28.613
L, light-gray portrait	60.00	49.676	29.806
L, dark-gray profile	58.50	48.185	28.911
Misa Amane, smiling	37.75	27.560	16.536
Misa Amane, choker	32.00	21.844	13.106
Near	20.33	10.244	6.146
Mello	40.00	29.796	17.878
Total	$B = 366.58$	284.669	$R = 170.801$

5.2 Draw-Until-Complete Baseline

For $n = 8$, $H_8 \approx 2.7179$, and Eq. (4.4) gives

$$\mathbb{E}[T] = 8H_8 \approx 21.7429.$$

Equation (4.5) gives expected net cost CNY 571.95 after resale recovery. Figure 3 shows that the distribution is strongly right-skewed: the set is eventually completed with probability one, but a small fraction of trials require many more draws than the mean.

More explicitly,

$$\mathbb{E}[C_{AR}] = 39.8(21.7429) - 170.801(2.717857 - 1) \approx 571.95 \text{ CNY}.$$

The final missing designs generate the right tail: new designs become increasingly rare as the collection approaches completion. Duplicate resale reduces net cost but does not eliminate this tail, because further draws continue to incur costs.

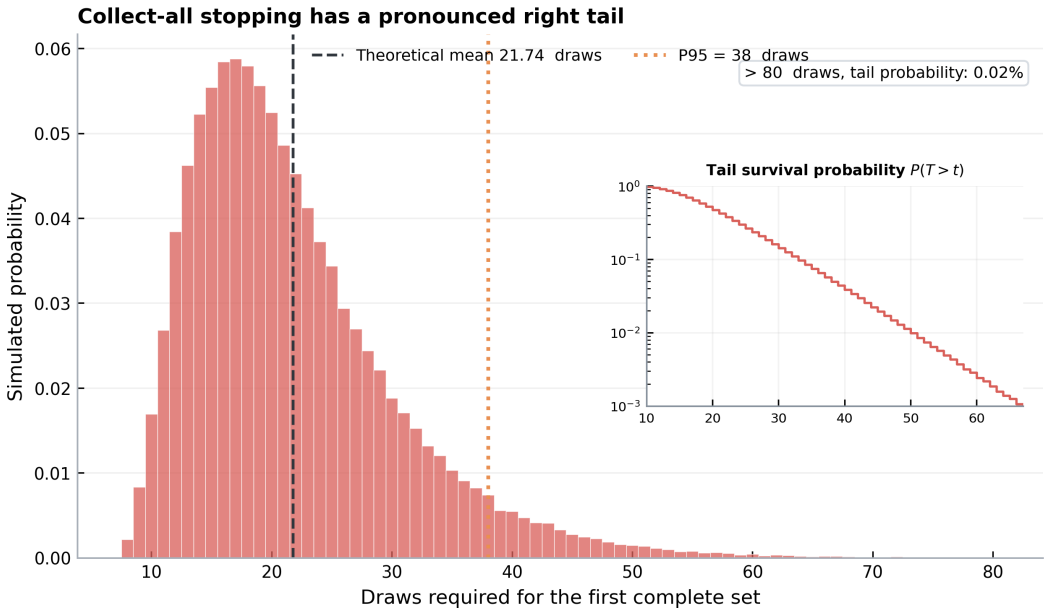


Figure 3. Main distribution and tail survival probability for the draw-until-complete strategy.

5.3 Fixed-Stop Strategy

Under the baseline parameters, the equal-probability expectation in Eq. (4.8) becomes

$$\mathbb{E}[C_{FR}(k)] = 39.8k + 366.58\left(\frac{7}{8}\right)^k - 170.801\left[\frac{k}{8} - 1 + \left(\frac{7}{8}\right)^k\right].$$

Table 5. Selected theoretical results for the fixed-stop strategy; cost components are in CNY.

k	Draw spending	Missing-item cost	Resale offset	Expected net cost
0	0.00	366.58	0.00	366.58
1	39.80	320.76	0.00	360.56
2	79.60	280.66	2.67	357.59
3	119.40	245.58	7.67	357.31
4	159.20	214.88	14.72	359.36
5	199.00	188.02	23.55	363.47

Table 5 gives selected values; the theoretical integer optimum is

$$k^* = 3, \qquad \mathbb{E}[C_{FR}(3)] = 357.3073 \text{ CNY}.$$

The value for $k = 2$ is about CNY 357.594, only CNY 0.287 higher. Expected cost therefore changes little between the two candidate limits. Figure 4 shows the near-tie and the associated price-bootstrap uncertainty.

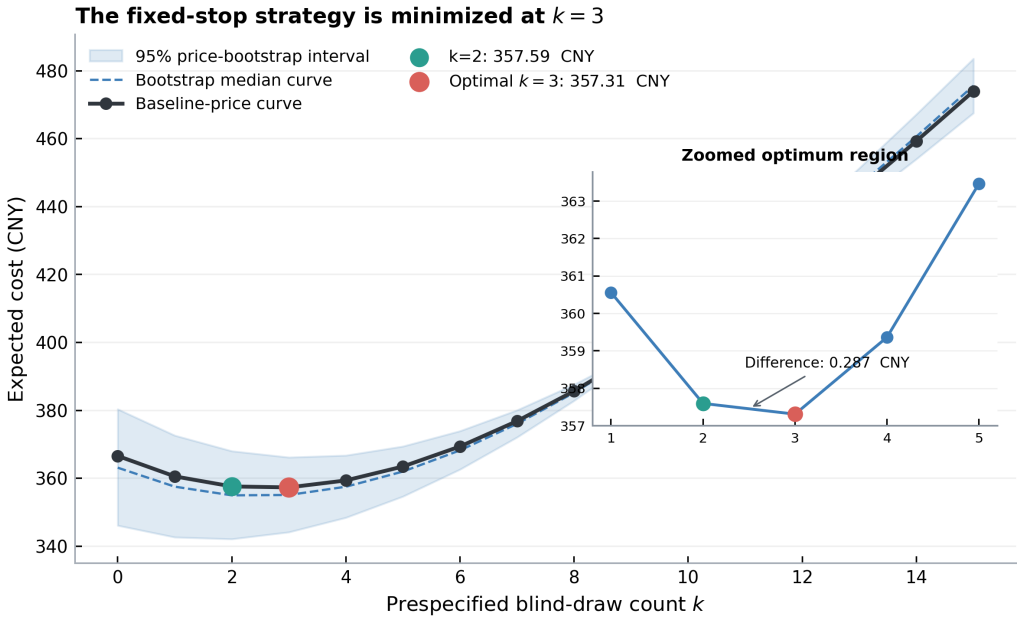


Figure 4. Theoretical expected cost of the fixed-stop strategy and its price-bootstrap interval.

The mechanism is visible from the adjacent difference Δ_k in Eq. (4.9). As k increases, expected completion purchases keep falling, but their marginal reduction decays geometrically. Duplicate counts and blind-draw spending continue to rise. Figure 5 compares the expected saving from one more draw with c: the saving remains slightly above the price after two draws but falls below it after the third. The optimum is therefore the crossing point of a declining marginal benefit and a constant marginal draw cost.

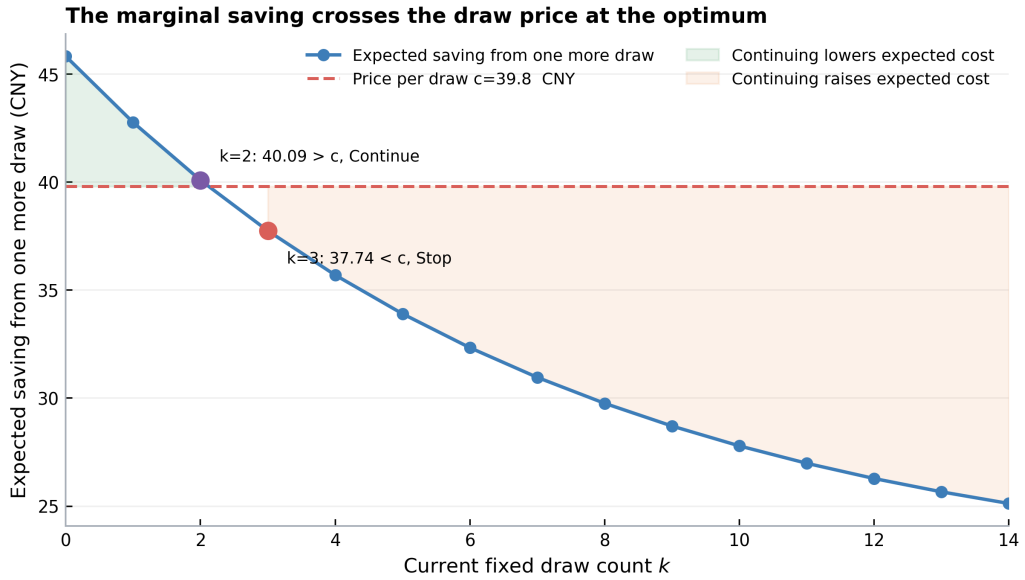


Figure 5. Intersection between the expected saving from one more draw and the blind-draw price.

5.4 Price-Aware Dynamic Stopping

The baseline state effects $\delta_i = b_i - r_i$ are approximately

$$30.19,\ 29.39,\ 30.19,\ 29.59,\ 21.21,\ 18.89,\ 14.18,\ 22.12.$$

The largest δ_i is below the 48.18 threshold in Eq. (4.13), so the policy continues after any first distinct design. In contrast, the three smallest effects satisfy

$$14.184 + 18.894 + 21.214 \approx 54.291 > 48.18.$$

Because every δ_i is positive, any state with at least three acquired designs must stop. The only layer at which design identity can change the decision is therefore the set of states with exactly two distinct designs.

Figure 6 evaluates all $\binom{8}{2} = 28$ two-design states: 10 prescribe continuation and 18 prescribe stopping. For example, acquiring Light Yagami (front view) and L (light-gray portrait) crosses the threshold, so the policy stops. Acquiring Near and Misa Amane (choker) leaves the sum below the threshold, so the policy continues. Table 12 in Appendix C lists all 28 decisions.

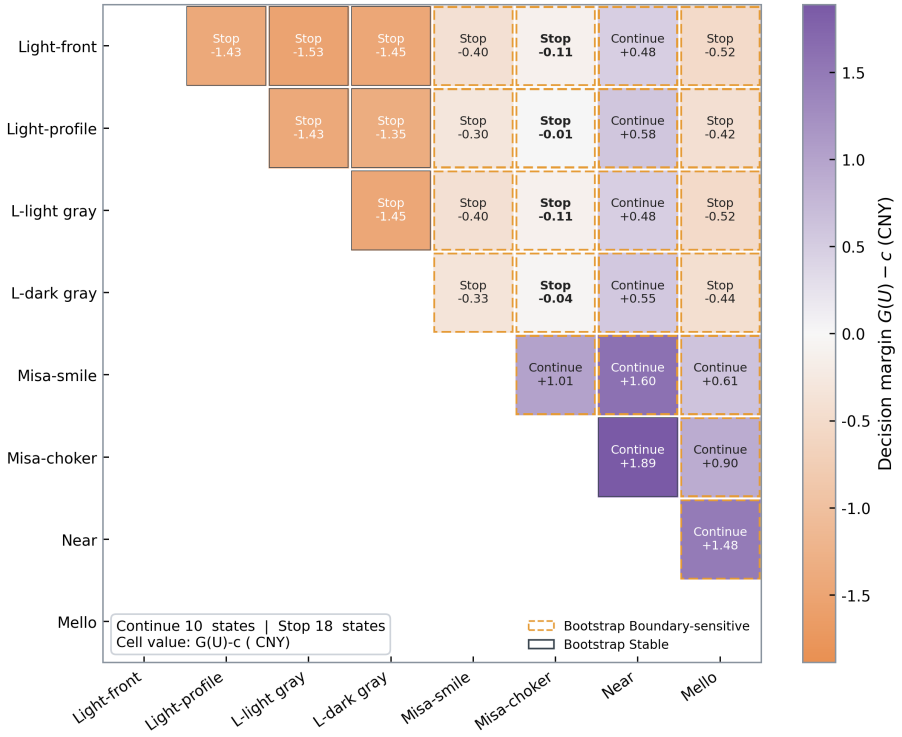


Figure 6. Actions and decision margins for all 28 states with two acquired designs.

Figure 7 shows how to apply the policy in Eq. (4.12) after each draw. The policy always continues from zero or one distinct design. When the second distinct design appears, the policy continues for 10 of the 28 possible pairs and stops for the other 18; if it continues, it stops when a third distinct design appears. Under the equal-probability sampling assumption, the policy stops almost surely and has a finite expected stopping time. Repeated draws can leave the state unchanged, so there is no deterministic upper bound on the total number of draws.

The 28 unordered pairs are equally likely when the second distinct design first appears. The policy therefore continues with probability $10/28$. The expected waiting time for the second distinct design is $8/7$ draws; conditional on continuation from a two-design state, the expected additional waiting time for the third is $8/6$ draws. Hence

$$\mathbb{E}[\tau_G] = 1 + \frac{8}{7} + \frac{10}{28}\frac{8}{6} = 2.6190. \tag{5.1}$$

Exact recursive evaluation gives

$$\boxed{\mathbb{E}[C_{\text{dyn}}] = 356.7156 \text{ CNY}.} \tag{5.2}$$

The dynamic policy saves about CNY 0.59 relative to fixed $k = 3$. This small gain reflects the value of using design identities in the stopping decision.

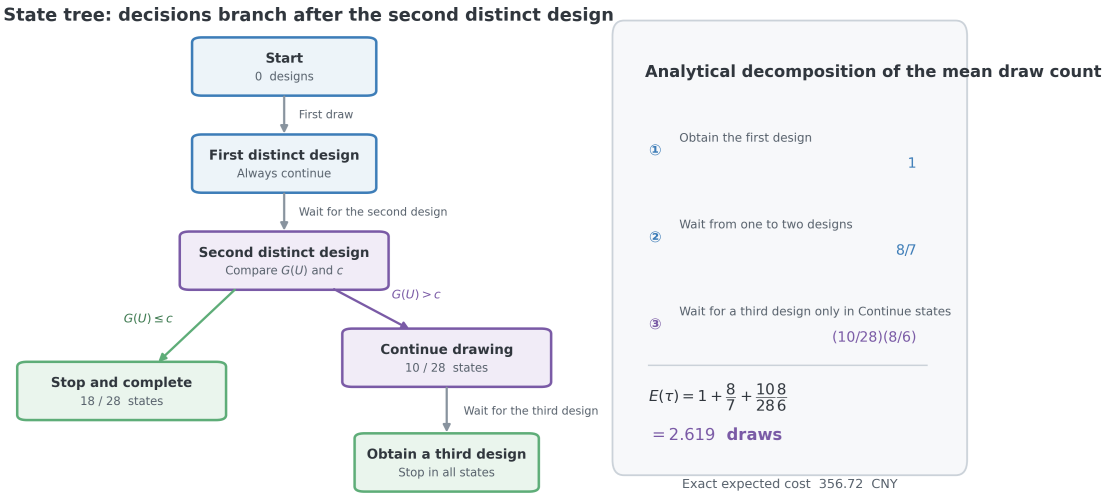


Figure 7. State evolution and action split under price-aware dynamic stopping.

5.5 Unified Expected-Cost Comparison

Table 6. Theoretical expected costs of the main strategies.

Strategy	Expected draws	Expected net cost (CNY)
Secondary market only	0	366.58
Fixed $k = 2$	2	357.59
Fixed $k = 3$	3	357.31
Price-aware dynamic stopping	2.619	356.72
Draw until complete	21.743	571.95

Table 6 compares the direct-purchase baseline in Eq. (4.3) with the stopping strategies. Dynamic stopping has the lowest baseline expected cost, given in Eq. (5.2), followed by fixed $k = 3$ and fixed $k = 2$. The small differences between these strategies motivate the risk and price-uncertainty comparisons below.

5.6 Analytical and Numerical Bellman Verification

Numerical agreement over 256 states is useful, but it does not explain why the one-step rule and the full dynamic program coincide. We therefore establish the equivalence analytically before using enumeration as a computational check. Recall the continuation cost in Eq. (4.15):

$$q(U) = \sum_{i \in U} p_i, \qquad C^*_{\text{cont}}(U) = \frac{c + \sum_{i \in U} p_i V^*(U \setminus \{i\}) - \sum_{j \notin U} p_j r_j}{q(U)}.$$

5.6.1 Analytical Meaning of the One-Step Condition

The optimal value can never exceed the immediate completion cost defined in Eq. (4.10). Hence, for every $i \in U$,

$$V^*(U \setminus \{i\}) \le B(U \setminus \{i\}) = B(U) - b_i.$$

Substituting these upper bounds into the continuation cost gives

$$\begin{aligned} C^*_{\text{cont}}(U) &\le \frac{c + q(U)B(U) - \sum_{i\in U} p_i b_i - \sum_{j\notin U} p_j r_j}{q(U)} \\ &= B(U) + \frac{c - G(U)}{q(U)}. \end{aligned}$$

Therefore, whenever $G(U) > c$,

$$C^*_{\text{cont}}(U) < B(U),$$

and the Bellman-optimal action must be Continue. Thus $G(U) > c$ is a sufficient continuation condition in every nonterminal state; this conclusion does not rely on enumerating future paths.

The reverse direction requires care. If $G(U) \le c$, later states might still contain valuable continuation options, so one-step reasoning alone does not generally prove that the current state should stop. However, if every one-step successor $U \setminus \{i\}$ is already known to stop, then

$$V^*(U \setminus \{i\}) = B(U \setminus \{i\}),$$

and the previous inequality becomes the exact identity

$$\boxed{C^*_{\text{cont}}(U) - B(U) = \frac{c - G(U)}{q(U)}.} \tag{5.3}$$

Under this successor condition, $G(U) > c$ is both necessary and sufficient for Bellman continuation; equality is assigned to Stop under the stated convention.

5.6.2 Monotonicity and the General Stopping Region

The successor condition can be proved without equal draw probabilities. Suppose that the probabilities are fixed and positive, purchase and expected recovery values are fixed, $b_i \ge r_i$ for every design, and $c > \sum_i p_i r_i$. The last condition rules out an expected profit from repeatedly drawing only duplicates and is satisfied by the baseline and all probability scenarios below. Completion is terminal, as in the model already defined.

Acquiring a previously missing design i changes the next-draw saving by

$$\boxed{G(U \setminus \{i\}) = G(U) - p_i(b_i - r_i) \le G(U).} \tag{5.4}$$

Equation (5.4) shows that the region $\{U : G(U) \le c\}$ is closed under acquiring additional designs. Once a state enters this region, no successor can restore a positive one-step margin.

We now induct on the number of missing designs. The empty set stops by definition and satisfies

$G(\varnothing) = \sum_i p_i r_i < c$. Assume that the claim has been established for all smaller missing sets. If a nonempty state has $G(U) \le c$, monotonicity places each $U \setminus \{i\}$ in the same stopping region. By the induction hypothesis its optimal value equals its immediate purchase cost. Substitution into the continuation identity in Eq. (5.3) gives

$$C^*_{\text{cont}}(U) - B(U) = \frac{c - G(U)}{q(U)} \ge 0,$$

so stopping is optimal. If instead $G(U) > c$, the upper-bound argument in the preceding subsection proves that continuing is strictly better than stopping. Equality uses the stated Stop convention. This establishes the threshold rule for every state under these conditions, including unequal-probability draws.

Positive probabilities ensure finite expected time to collect all designs, so the policies considered here have finite expected stopping times when completion is terminal. The proof concerns additive expected cost. Delayed payments, price changes, or a risk-constrained objective can require additional state variables and do not follow from this threshold argument.

5.6.3 Backward Induction for the Baseline Case

Under equal probabilities,

$$G(U) = \frac{1}{8}\left(B - \sum_{i \in O} \delta_i\right), \qquad G(U) > c \Longleftrightarrow \sum_{i \in O} \delta_i < 48.18.$$

The terminal state $|O| = 8$ stops by definition. Since the sum of the three smallest δ_i values is about 54.291, every state with $|O| \ge 3$ has $G(U) < c$. Starting at $|O| = 7$ and working backward to $|O| = 3$, every successor lies in a layer already proved to stop. Equation (5.3) therefore applies at each layer and shows that all these states stop.

When $|O| = 2$, every successor has three acquired designs and has just been proved to stop. The action comparison in Eq. (5.3) again reduces exactly to

$$\delta_i + \delta_j < 48.18.$$

This produces the same 10 Continue and 18 Stop states shown in Figure 6. When $|O| = 1$, the largest single effect is about 30.19, so

$$G(U) \ge \frac{366.58 - 30.19}{8} \approx 42.05 > 39.8 = c.$$

At the initial state, $G(U_0) = 366.58/8 \approx 45.82 > c$. The sufficient condition therefore proves Continue for the zero- and one-design layers. Together, these arguments cover every state analytically.

5.6.4 Exhaustive State Check

The program independently evaluates the threshold action and the Bellman recursion in Eq. (4.16) for all $2^8 = 256$ missing-item sets. It obtains Continue at the initial state, Continue in all eight one-design states, a 10–18 split across the 28 two-design states, and Stop after at least three distinct designs. The conflict count is zero. The Bellman initial value is

$$V^*(U_0) = 356.7156 \text{ CNY},$$

which equals the exact value of the price-aware rule. The equality follows from the layer-by-layer proof: both methods select the same action and obey the same value recursion in every reachable state. It is conditional on fixed probabilities, fixed prices, and additive expected recovery; dynamic prices, hard budgets, or nonlinear risk preferences would change the state and value structure.

5.7 Monte Carlo Risk Analysis

Simulation follows standard independent-replication practice [13, 14]; Appendix A gives the procedures. We use 100,000 trials for each randomized strategy. A fixed policy completes its prescribed k draws and then buys missing items; the dynamic policy recomputes $G(U)$ before paying for each additional draw. Each duplicate resale is generated as an explicit Bernoulli(η_i) event rather than replaced by its expectation. Independent, fixed pseudorandom-number streams are used across strategies.

Table 7 reports the mean, standard deviation, median, 90th and 95th percentiles (P90 and P95), and probability that final net cost C exceeds CNY 400. Quantiles use the linear sample-quantile convention discussed by Hyndman and Fan [15]. The CNY 400 line is not claimed to be a universal personal budget; it is an interpretable integer reference about 9.1% above the deterministic secondary-market cost. Appendix B therefore also reports the full net-cost exceedance curve across alternative thresholds.

Table 7. Monte Carlo risk summary; monetary statistics are in CNY.

Strategy	Mean	SD	Median	P90	P95	$\mathbb{P}(C > 400)$
Secondary market only	366.58	0.00	366.58	366.58	366.58	0.00%
Fixed $k = 2$	357.60	22.09	354.18	388.10	393.85	3.49%
Fixed $k = 3$	357.24	28.26	355.97	395.48	408.23	7.27%
Dynamic stopping	356.81	25.95	349.93	393.65	407.45	6.20%
Draw until complete	572.60	179.78	537.09	809.89	916.38	86.02%

5.7.1 Purchase Outlay Before Resale

The same simulated draw sequences also give purchase outlay A as defined in Eq. (4.2), without subtracting duplicate recovery. Table 8 compares its exceedance probability with that of final net cost. These are paired measurements on the same trials, not simulations of a new stopping policy. Key analytical values are also collected in Appendix D.

Table 8. Purchase-outlay risk before resale; monetary statistics are in CNY.

Strategy	Mean A	P95 of A	$\mathbb{P}(A > 400)$	$\mathbb{P}(C > 400)$
Secondary market only	366.58	366.58	0.00%	0.00%
Fixed $k = 2$	360.23	406.18	6.29%	3.49%
Fixed $k = 3$	364.87	425.65	13.27%	7.27%
Dynamic stopping	361.74	427.70	9.93%	6.20%
Draw until complete	866.39	1512.40	97.26%	86.02%

A small exact enumeration makes the distinction especially clear. For fixed $k = 2$, we enumerate all

8^2 draw sequences and the possible Bernoulli resale outcomes, weighting each outcome by its probability. The resulting probabilities are

$$\mathbb{P}(C_{FR}(2) > 400) = 3.4375\%, \qquad \mathbb{P}(A_{FR}(2) > 400) = 6.25\%.$$

The first agrees with the Monte Carlo estimate of 3.49% within simulation error. The second is larger because potential resale income cannot pay for earlier purchases under the delayed-receipt scenario. Neither probability guarantees that a collector starting with CNY 400 can complete the set: seller expenses, settlement timing, and actual cash availability have not been modeled. A hard-budget optimal policy would need a cash-balance state and an explicit feasibility constraint.

5.7.2 Net-Cost Tail Comparison

Figures 8 and 9 show that a lower mean does not guarantee a lower upper tail. Fixed $k = 2$ sacrifices less than CNY 0.30 in theoretical mean cost relative to $k = 3$ but cuts the estimated probability that final net cost exceeds CNY 400 by more than half.

This separation remains after accounting for Monte Carlo uncertainty. Wilson 95% intervals for exceedance are $[3.38\%, 3.61\%]$ for fixed $k = 2$, $[7.11\%, 7.43\%]$ for fixed $k = 3$, and $[6.06\%, 6.36\%]$ for dynamic stopping [16, 17]. Approximate order-statistic intervals for P95 are $[406.95, 407.70]$ CNY for the dynamic policy and $[407.65, 408.23]$ CNY for fixed $k = 3$. The empirical distribution of fixed $k = 2$ has discrete mass near its 95th percentile, so both endpoints of its reported interval equal CNY 393.85; this reflects a finite set of cost outcomes rather than zero sampling uncertainty.

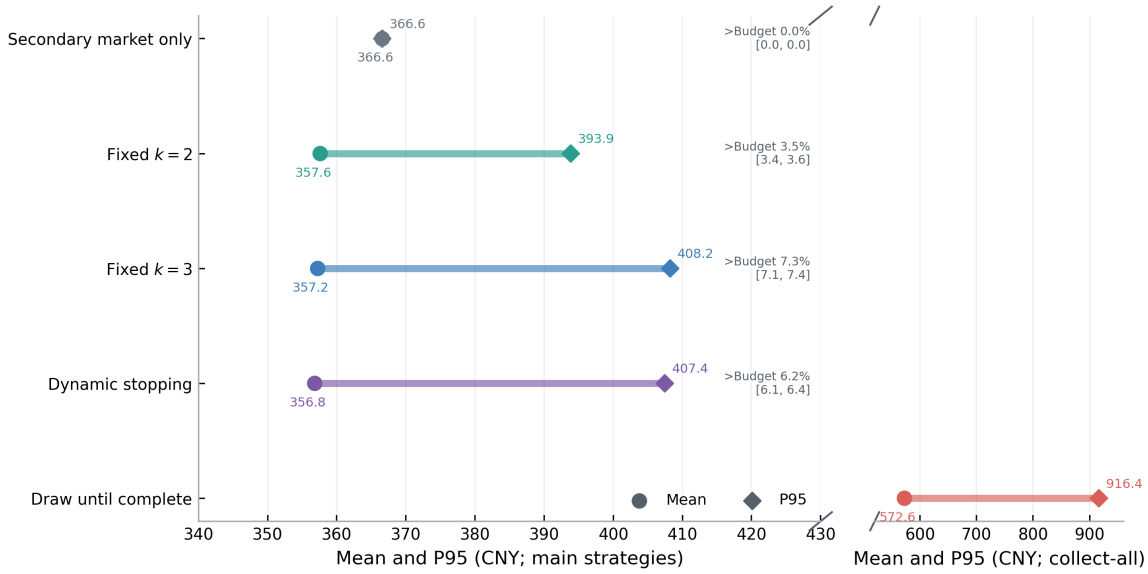


Figure 8. Mean and P95 comparison across the five strategies.

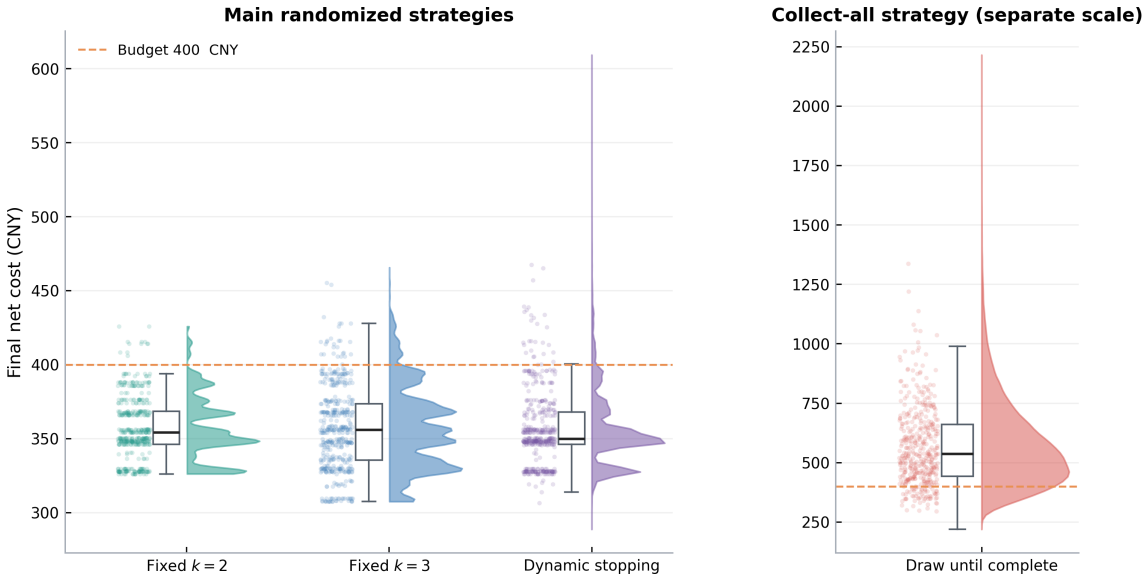


Figure 9. Distribution of final net cost for the four randomized strategies.

5.8 Agreement Between Theory and Simulation

The simulated means are close to the corresponding analytical or recursive expectations. The dynamic mean is CNY 356.81, about CNY 0.10 above its exact value of CNY 356.72. The fixed-$k = 3$ mean is CNY 357.24, about CNY 0.07 below its theoretical value of CNY 357.31. Similar agreement holds for secondary-market-only, fixed $k = 2$, and draw-until-complete strategies.

Appendix Figure 21 plots "Monte Carlo mean minus theoretical expectation" together with 95% Monte Carlo confidence intervals. The theoretical expectations fall within the corresponding 95% Monte Carlo intervals. This supports the consistency of the closed-form calculations, exact state recursion, and simulation.

5.9 Risk-Based Recommendations

The choice depends on the collector's objective:

(1) **Lowest expected cost:** dynamic stopping gives exact expected net cost CNY 356.72; fixed $k = 3$ is the best prespecified limit under baseline inputs.

(2) **Lower final-net-cost risk:** fixed $k = 2$ has P95 of CNY 393.85 and an estimated 3.49% probability of exceeding CNY 400. Its purchase-outlay risk is higher, so this is not a cash-budget guarantee.

(3) **Avoiding blind draws and resale:** direct purchase costs CNY 366.58 under the representative prices and requires no resale management.

Drawing until completion is unsuitable as a money-saving rule in this scenario: its mean cost and upper tail are substantially higher.

6 Sensitivity Analysis and Model Discussion

Sensitivity analysis separates an exact optimum under the baseline scenario from conclusions that remain stable under parameter perturbation [18]. We first vary individual decision parameters and examine the interaction between draw cost and resale success. We then perturb draw probabilities, propagate finite-sample price uncertainty through a nonparametric bootstrap, and check the effect of the flagged price records.

6.1 Effect of Resale Success Probability

Let the sum of the per-design net revenues from successful sales be

$$G_{\text{sale}} = \sum_i g_i \approx 284.669, \qquad R(\eta) = \eta G_{\text{sale}}.$$

The switch between fixed $k = 2$ and $k = 3$ occurs when their expected costs are equal. Substitution into Eq. (4.8) gives

$$\eta_{2\to3} \approx 0.5656.$$

When comparing these two fixed limits, $k = 2$ has the lower expected cost below about 56.6% resale success, while $k = 3$ is preferable above that threshold. Because the baseline $\eta = 0.60$ is close to this boundary, the two fixed costs are nearly tied. Re-optimization at representative values gives

$$\eta = 0.30, 0.40, 0.50 \Rightarrow k^* = 2, \quad \eta = 0.60, 0.70 \Rightarrow k^* = 3, \quad \eta = 0.80 \Rightarrow k^* = 4.$$

Improved resale conditions support additional draws because duplicates lose less economic value.

6.2 Effect of the Secondary-Market Price Level

Let λ scale all secondary-market prices, $b_i(\lambda) = \lambda b_i$, and recompute $g_i(\lambda)$ and $r_i(\lambda)$ under the same accounting rule. The boundary between zero and one blind draw is especially transparent: the first draw cannot be a duplicate, so its expected avoided purchase cost is $\lambda B/n$. The switch occurs at

$$\lambda_c = \frac{nc}{B} = \frac{8(39.8)}{366.58} \approx 0.8686.$$

Below roughly 86.9% of the observed price level, even the first blind draw is unattractive in expectation, so direct market purchase may be optimal. For $\lambda = 0.8, 0.9, 1.0, 1.1, 1.2$, the optimized fixed counts are $k^* = 0, 1, 3, 4, 6$, respectively. Higher completion prices increase the value of blind draws because avoiding a missing-item purchase becomes more valuable.

6.3 Effect of Blind-Draw Cost

Table 9 shows the optimal fixed draw count with the remaining parameters held fixed:

Table 9. Optimal fixed draw count as the blind-draw price changes.

c (CNY)	35.0	37.5	39.8	42.5	45.0
k^*	5	4	3	2	1

As c rises, the collector switches earlier to secondary-market completion. The same direction appears in the dynamic model: increasing c reduces the number of two-design states for which $G(U) > c$.

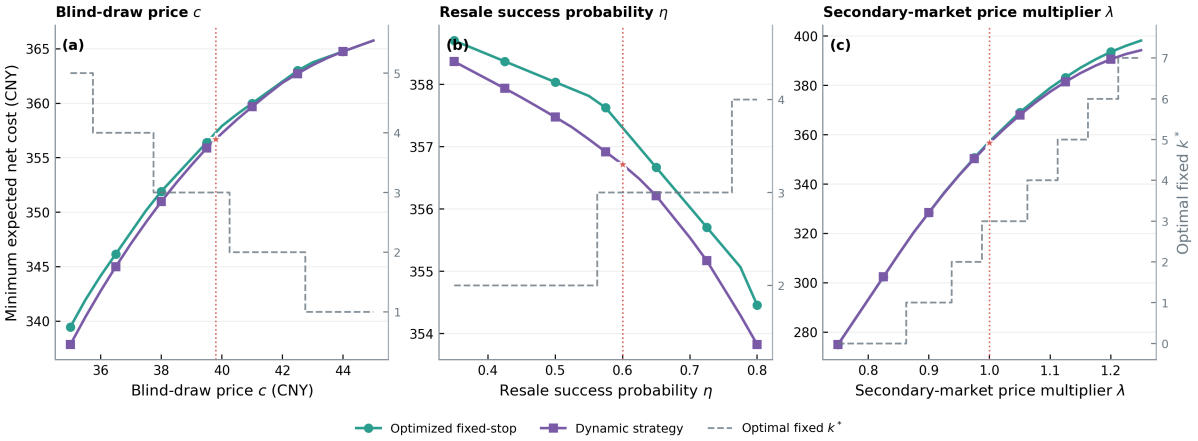


Figure 10. One-way sensitivity of optimized fixed stopping and dynamic stopping to c, η, and λ.

Figure 10 compares the cost responses in the three one-way analyses. Higher c raises both optimized costs and reduces fixed k^* stepwise; higher η lowers net cost and delays stopping; higher λ raises completion cost and makes additional draws more valuable. Across the scanned ranges, dynamic stopping has expected cost no higher than optimized fixed stopping. Using the observed designs allows the dynamic policy to adjust its decisions after each draw.

6.4 Transaction Frictions

Platform fees, buyer-side shipping adjustments, and seller costs all enter through

$$g_i = \max\{(1-\gamma)(b_i - d) - h, 0\}.$$

We vary $\gamma \in [0, 0.05]$, $d \in [0, 12]$, and $h \in [0, 10]$ one at a time and re-solve both stopping models at every grid point. Increasing these costs can reduce duplicate recovery, raise expected net cost, and favor earlier stopping.

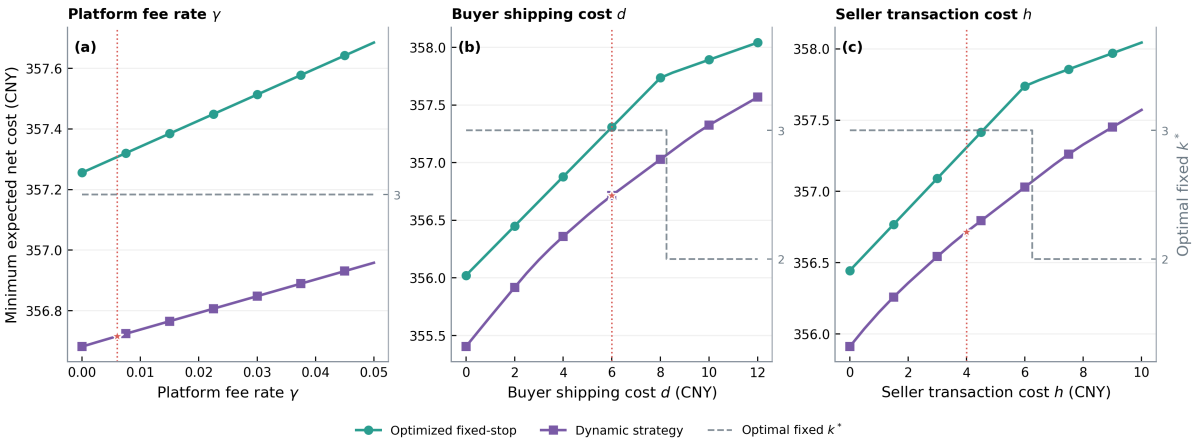


Figure 11. One-way sensitivity to the platform fee rate γ, shipping adjustment d, and seller cost h.

The directions are consistent, but their magnitudes differ. Over the selected ranges, γ has a smaller effect because it removes only a percentage of the transaction base, whereas d and h directly subtract currency amounts from every successful sale. Resale probability η and transaction frictions also have distinct meanings: η governs whether a duplicate sells, while γ, d, and h govern how much remains after a successful sale.

6.5 Joint Sensitivity of Draw Price and Resale Success

One-way analysis cannot show how two parameters interact. Figure 12 therefore recomputes the optimal fixed k and the number of two-design Continue states over the (c, η) grid. Low draw price and high resale success jointly increase k^* and expand the dynamic continuation region. High draw price and low resale success contract both. The fixed model moves through discrete integer regions, whereas the dynamic model adjusts state by state. Thus neither k^* nor the dynamic boundary is a permanent property of the product; each depends on both purchasing cost and the resale environment.

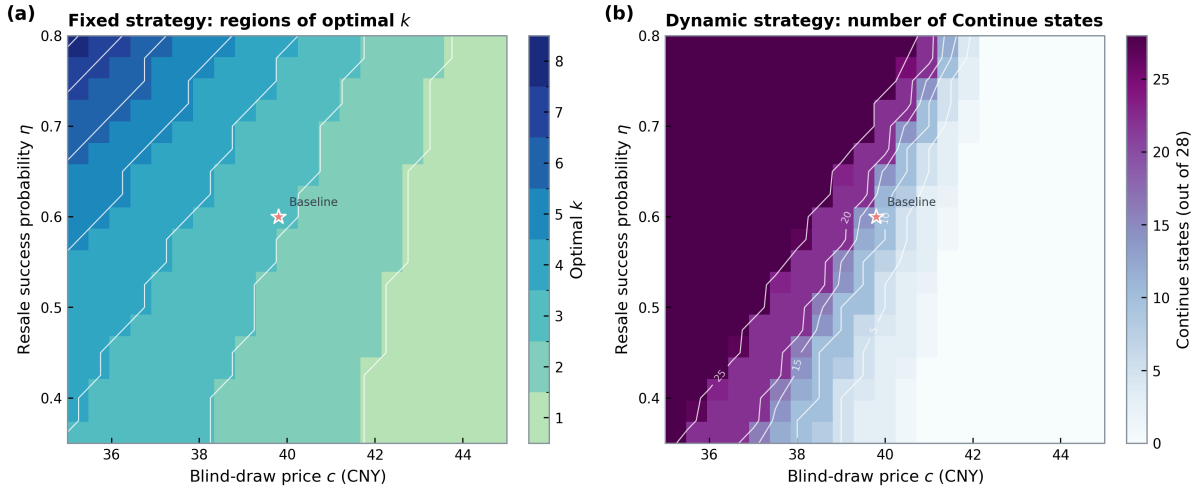


Figure 12. Joint sensitivity of the optimal fixed draw count and the dynamic continuation region to c and η.

6.6 Unequal Draw Probabilities

Mixing loose packs does not establish exact equality of design probabilities. To examine a limited departure from the baseline, we shift probability between the four higher-priced Light/L designs and the other four designs:

$$p_i(\delta) = \begin{cases} 1/8 + \delta, & i = 1, 2, 3, 4, \\ 1/8 - \delta, & i = 5, 6, 7, 8, \end{cases} \qquad -0.025 \le \delta \le 0.025.$$

This construction preserves positive probabilities and a total of one. A positive δ makes the higher-priced designs more common; a negative value makes them less common. These are constructed stress scenarios, not estimates of the actual product probabilities. They examine one relevant direction of uncertainty rather than all possible probability vectors.

For each of 51 equally spaced values of δ, we recompute the general fixed-stop expectation in Eq. (4.7), the dynamic value, and all Bellman state actions. The conditions of the monotonicity proof hold throughout this range. All 51 scenarios have zero action conflicts between the threshold and Bellman rules. Table 10 reports the endpoints and baseline; p_H and p_L are the per-design probabilities for the first and second groups.

Table 10. Selected unequal-probability scenarios, with all other inputs held fixed.

p_H	p_L	Fixed k^*	Fixed cost (CNY)	Dynamic cost (CNY)
0.100	0.150	2	362.77	362.70
0.125	0.125	3	357.31	356.72
0.150	0.100	3	350.40	349.08

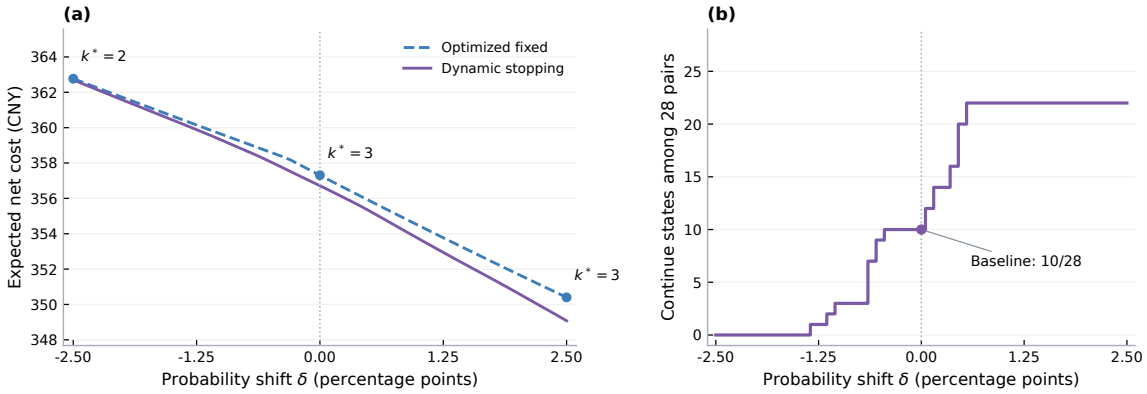


Figure 13. Probability perturbations change expected costs and the two-design continuation region. Panel (a) also marks the optimal fixed count at the endpoints and baseline.

When the higher-priced designs become less common, blind draws avoid fewer expensive purchases and the optimal fixed count falls to two at the lower endpoint. Increasing their probabilities lowers the optimized expected costs, although the fixed optimum at the upper endpoint remains three. Figure 13 also shows that the baseline 10–18 dynamic split is not invariant. The mathematical threshold rule remains valid under the stated conditions, but its numerical recommendations must use the appropriate probability vector.

6.7 Price-Sample Uncertainty

We use 2,000 within-design nonparametric bootstrap resamples to propagate finite price-sample uncertainty [19]. Each resample recalculates the eight medians, the optimized fixed strategy, and every dynamic state action. This procedure approximates the sampling distribution without imposing a parametric form on listing prices, which is useful when each design has only 13–20 usable observations.

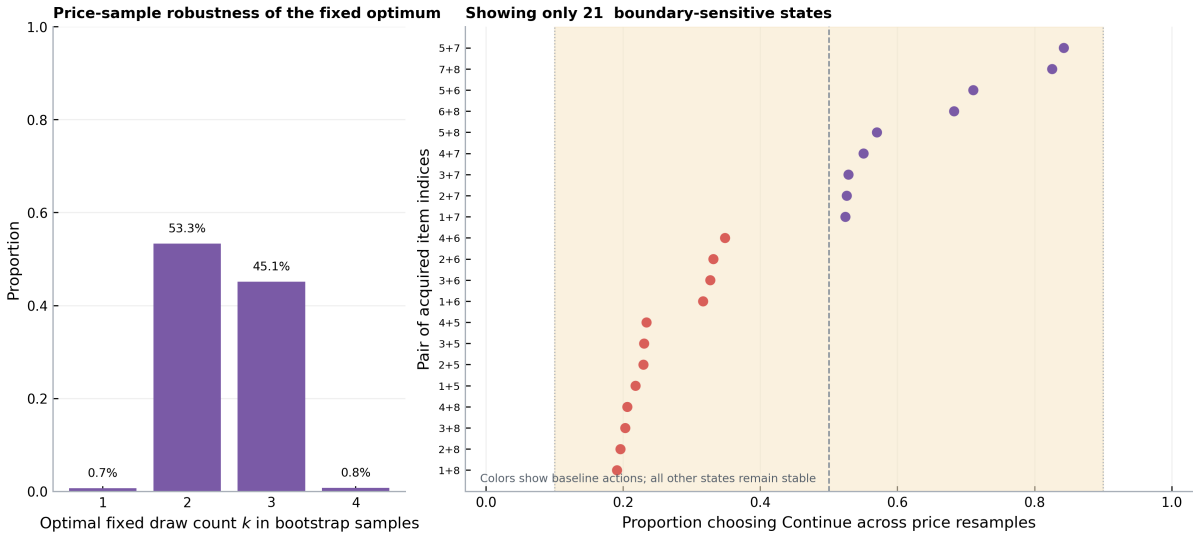


Figure 14. Bootstrap robustness of the fixed optimum and the dynamic two-design state boundary.

Figure 14 shows that the fixed optimum is $k = 2$ in 53.35% of resamples and $k = 3$ in 45.15%; $k = 1$ and $k = 4$ together account for only 1.50%. The baseline medians make $k = 3$ exactly optimal, but the bootstrap selects either $k = 2$ or $k = 3$ in most resamples. A CNY 0.29 baseline difference should therefore not be interpreted as a stable universal advantage.

For dynamic decisions, a state is called stable Stop if its Continue proportion is at most 0.10, stable Continue if the proportion is at least 0.90, and boundary-sensitive otherwise. Among the 28 two-design states, six are stable Stop, one is stable Continue, and 21 are boundary-sensitive. This does not contradict the exact 10–18 split at the baseline medians. It shows that many decision margins are comparable to the sampling uncertainty in prices. For a boundary-sensitive state, the stopping decision should be recalculated using current prices. The row-level bootstrap is conditional on treating the retained price records as resampling units; shared listings or sellers can create dependence and reduce the effective information in the sample.

6.8 Conservative Check of Flagged Price Records

To assess the unresolved records without rewriting their original fields, we perform a separate conservative check. It excludes both MKT-068 and MKT-075, which share a design and a source URL, and MKT-126, whose year lies outside the main observation window. Removing both linked records avoids choosing which one to retain without further evidence. This leaves 122 observations. The median for the light-gray L design falls from CNY 60 to CNY 57.50, and the Mello median falls from CNY 40 to CNY 39; the other six medians are unchanged.

Table 11. Expected-cost sensitivity to excluding the three flagged records; costs are in CNY.

Records	Direct cost	Fixed k^*	Fixed cost	Dynamic cost
125	366.58	3	357.31	356.72
122	363.08	2	354.95	354.32

Table 11 reports recovery values and expected costs recalculated from each set of medians. Costs are in CNY and all nonprice inputs remain fixed. The fixed optimum changes from three draws to two; the dynamic rule still has a slightly lower expected cost than the optimized fixed rule. This is a deterministic data-quality check, separate from the 2,000-resample bootstrap. It does not establish that the remaining observations are independent or representative, and it does not resolve the missing search and seller information.

6.9 Shared Marginal Principle of Fixed and Dynamic Rules

Both strategies compare the expected saving from another draw with its price. The fixed rule compares the expected saving from draw $k+1$ with its price; the dynamic rule compares $G(U)$ with c. Increasing c makes another draw less attractive in both models. Increasing η or reducing transaction frictions raises r_i, improves the value of duplicates, and supports continued drawing.

Near the baseline, Figures 10 and 12 show that raising c from CNY 35 to CNY 45 reduces the fixed optimum from five draws to one and contracts the two-design dynamic Continue region toward zero. Figure 11 shows the same early-stopping effect when duplicate recovery deteriorates. A higher draw price or lower duplicate recovery makes continued drawing less attractive in both models.

6.10 Sampling-Mechanism Boundary

A verified complete box containing each design exactly once is not an independent-draw process. If k packs are selected uniformly from that box at the same unit price, each design is acquired with probability $k/8$. The selected designs are distinct, the expected remaining purchase cost is $(1-k/8)B$, and there are no duplicates to resell. Therefore

$$\mathbb{E}[C_{\text{box}}(k)] = kc + \left(1 - \frac{k}{8}\right) B, \qquad k = 0, 1, \ldots, 8. \tag{6.1}$$

Because $c = 39.8 < B/8 \approx 45.82$, the expected cost in Eq. (6.1) decreases with each additional item purchased from the verified box. The optimum is $k = 8$, with cost $8c = 318.40$ CNY, about CNY 38.32 below the loose-pack dynamic policy. This result requires availability of a complete box, the same per-item cost, and a guaranteed nonrepeating composition. It cannot be transferred to mixed loose packs from different sources. Figure 15 is therefore a model-boundary check: the sampling mechanism must be identified before any stopping strategy is applied.

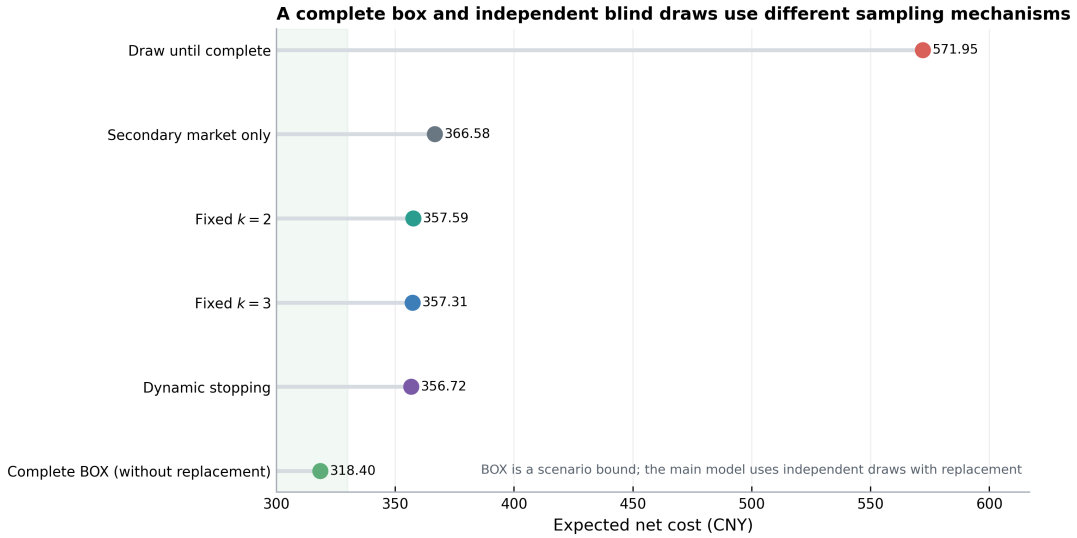


Figure 15. Expected-cost comparison between independent loose-pack strategies and a nonrepeating complete-box scenario.

6.11 Model Strengths

(1) **Consistent cost accounting.** All strategies use the same purchase and recovery convention, allowing analytical expectations and simulated costs to be compared directly.

(2) **An interpretable stopping rule.** The dynamic decision requires the missing-design set and a comparison of $G(U)$ with c. The Bellman proof establishes optimality under the stated conditions, with enumeration providing an independent action check.

(3) **Explicit decision boundaries.** Sensitivity analysis and price resampling show when the fixed optimum or dynamic state action changes, rather than treating the baseline recommendation as permanent.

6.12 Model Limitations

(1) **Sampling approximation.** Independent equal-probability sampling represents a mixed loose-pack scenario, not a measured loose-pack probability distribution. The constructed perturbations test one form of unequal probability but do not calibrate the true distribution. Purchases from one box may instead be without replacement; extending the collect-all formulas to unequal probabilities or grouped draws requires generalized Coupon Collector models [3, 4].

(2) **Short-run prices.** The records are concentrated in August 2026, with one unresolved date year. Their medians describe a market snapshot rather than forecast future values.

(3) **Simplified resale mapping.** The parameters η, d, and h are scenario inputs rather than estimates from completed transaction histories. Transaction prices, time to sale, and design-specific sale probabilities would support richer estimates.

(4) **Finite and selected price samples.** The bootstrap describes within-sample variation but cannot correct platform-selection bias, listing-to-transaction price differences, or unrecorded seller dependence.

The conservative record check tests the identified anomalies without claiming that all source evidence has been verified.

(5) **Unmodeled nonmonetary utility.** Entertainment, emotional value, trading enjoyment, and waiting time may matter to collectors but are outside the monetary objective.

(6) **Risk-neutral dynamic objective.** The current Bellman model minimizes expected net cost. It reports purchase outlay separately but does not impose a cash constraint or model settlement times. Direct optimization of P95, exceedance probability, hard-budget feasibility, or risk-averse utility would require an expanded state and objective.

7 Conclusion

For this eight-design collection, limited blind draws followed by secondary-market completion have lower expected net cost than drawing until completion under the independent, equal-probability loose-pack model. At the baseline inputs, drawing until completion requires 21.74 draws on average and costs CNY 571.95 after expected duplicate recovery. Fixed $k = 3$ costs CNY 357.31, only CNY 0.29 below fixed $k = 2$. Dynamic stopping lowers the expected cost to CNY 356.72 and uses 2.619 draws on average. It continues after zero or one distinct design, continues in 10 of the 28 two-design states, and stops after at least three distinct designs. Under the stationary, additive conditions proved in Section 5.6, the one-step threshold and Bellman model choose the same action; all 256 baseline states and all 51 probability scenarios have zero action conflicts.

The lowest mean cost does not imply the lowest tail risk. Fixed $k = 2$ has P95 of CNY 393.85 and an estimated 3.49% chance that final net cost exceeds CNY 400, compared with 7.27% for fixed $k = 3$ and 6.20% for dynamic stopping. Its exact purchase-outlay exceedance probability is 6.25%, so the net-cost comparison is not a cash-budget guarantee. Price resampling selects $k = 2$ and $k = 3$ in 53.35% and 45.15% of resamples and identifies 21 boundary-sensitive two-design states. These small mean-cost differences should be assessed alongside risk and current price information.

The recommended strategy depends on prices, resale conditions, and how packs are sampled. The fixed optimum switches between two and three draws near a resale success probability of 56.6%; direct purchase may become preferable below about 86.9% of the baseline price level. Higher draw costs favor earlier stopping. If a verified nonrepeating complete box is available at the same unit cost, purchasing all eight designs costs CNY 318.40 and changes the preferred strategy. The listing sample and scenario resale inputs limit how far these results can be generalized. A collector should first identify the sampling mechanism, then apply the stopping rule using updated purchase prices and recovery conditions. Hard-budget or risk-averse decisions require a different objective and additional state information.

References

[1] W. Feller. *An Introduction to Probability Theory and Its Applications, Volume I*. 3rd ed. Wiley, 1968.

[2] A. Boneh and M. Hofri. "The Coupon-Collector Problem Revisited—A Survey of Engineering Problems and Computational Methods". In: *Communications in Statistics. Stochastic Models* 13.1 (1997), pp. 39–66. DOI: 10.1080/15326349708807412.

[3] E. Anceaume, Y. Busnel, and B. Sericola. "New Results on a Generalized Coupon Collector Problem Using Markov Chains". In: *Journal of Applied Probability* 52.2 (2015), pp. 405–418. DOI: 10.1239/jap/1437658606.

[4] W. Stadje. "The Collector's Problem with Group Drawings". In: *Advances in Applied Probability* 22.4 (1990), pp. 866–882. DOI: 10.2307/1427566.

[5] M. Ferrante and M. Saltalamacchia. "The Coupon Collector's Problem". In: *MATerials MATemàtics* 2014.2 (2014), pp. 1–35. URL: https://mat.uab.cat/~matmat/PDFv2014/v2014n02.pdf (visited on 10/15/2026).

[6] R. Bellman. *Dynamic Programming*. Princeton, NJ: Princeton University Press, 1957.

[7] M. L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. New York: John Wiley & Sons, 1994. DOI: 10.1002/9780470316887.

[8] PARCO Co., Ltd. *The First "DEATH NOTE -My Dear Light-" Pop-Up Shop Featuring Misa Amane*. May 8, 2025. URL: https://prtimes.jp/main/html/rd/p/000003340.000003639.html (visited on 10/15/2026).

[9] PARCO ART. *DEATH NOTE -My Dear Light- POPUP SHOP*. Historical event-detail link; it redirects to the PARCO ART homepage as of October 15, 2026. Product specifications are available in the PARCO press release. 2025. URL: https://art.parco.jp/nagoya/detail/?id=1648 (visited on 10/15/2026).

[10] Project Researcher. *Secondary-Market Price Records and Cleaning Table for the Eight "DEATH NOTE -My Dear Light-" Badge Designs*. Internal project dataset dated mainly August 5–19, 2026, including source URLs, reported screenshot filenames, and inclusion or exclusion labels. One date year remains unresolved; record-level mapping and audit notes are supplied with the computational project. 2026.

[11] P. J. Huber and E. M. Ronchetti. *Robust Statistics*. 2nd ed. John Wiley & Sons, 2009. DOI: 10.1002/9780470434697.

[12] S. M. Ross. *A First Course in Probability*. 10th ed. Pearson, 2019.

[13] R. Y. Rubinstein and D. P. Kroese. *Simulation and the Monte Carlo Method*. 3rd ed. Wiley, 2016. DOI: 10.1002/9781118631980.

[14] A. M. Law. *Simulation Modeling and Analysis*. 5th ed. McGraw-Hill Education, 2015.

[15] R. J. Hyndman and Y. Fan. "Sample Quantiles in Statistical Packages". In: *The American Statistician* 50.4 (1996), pp. 361–365. DOI: 10.1080/00031305.1996.10473566.

[16] E. B. Wilson. "Probable Inference, the Law of Succession, and Statistical Inference". In: *Journal of the American Statistical Association* 22.158 (1927), pp. 209–212. DOI: 10.1080/01621459.1927.10502953.

[17] R. G. Newcombe. "Two-Sided Confidence Intervals for the Single Proportion: Comparison of Seven Methods". In: *Statistics in Medicine* 17.8 (1998), pp. 857–872. DOI: 10.1002/(SICI)1097-0258(19980430)17:8<857::AID-SIM777>3.0.CO;2-E.

[18] A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, and S. Tarantola. *Global Sensitivity Analysis: The Primer*. John Wiley & Sons, 2008. DOI: 10.1002/9780470725184.

[19] B. Efron and R. Tibshirani. "Bootstrap Methods for Standard Errors, Confidence Intervals, and Other Measures of Statistical Accuracy". In: *Statistical Science* 1.1 (1986), pp. 54–75. DOI: 10.1214/ss/1177013815.

A Monte Carlo Simulation Procedure

For each randomized strategy, the simulation uses an independent pseudorandom-number stream derived from the base seed 20260917. Each trial records the draw sequence, missing-item purchases, duplicate resale events, final net cost, and stopping time. The principal experiment uses 100,000 trials per strategy.

A.1 Fixed-Stop Strategy

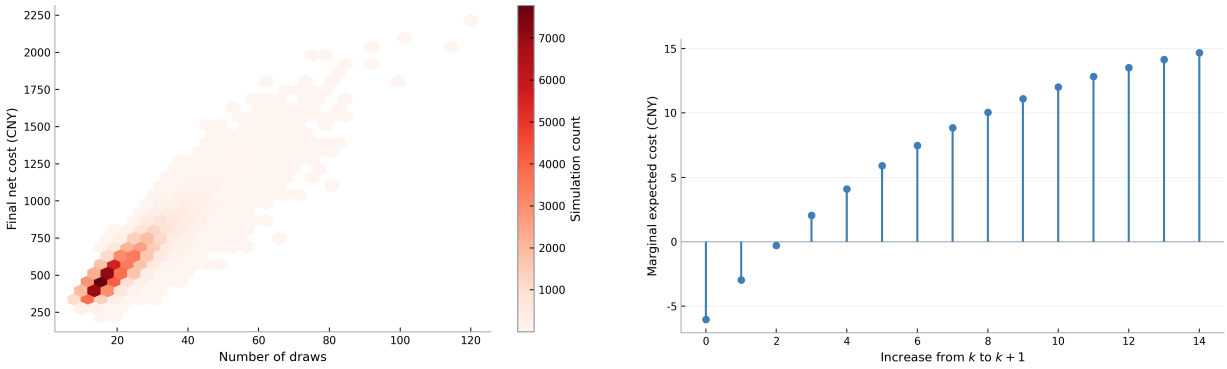
1. Initialize the occurrence counts of all eight designs.
2. Generate exactly k independent draws according to $(p_1, \ldots, p_n)$.
3. Purchase every design whose occurrence count remains zero at cost b_i.
4. Treat every copy beyond the first as a duplicate.
5. For each duplicate of design i, generate an independent Bernoulli(η_i) sale outcome and subtract g_i when the sale succeeds.
6. Record the realized net cost in Eq. (4.6), and repeat for all trials.

A.2 Price-Aware Dynamic Strategy

1. Initialize the missing set as $U = \{1, \ldots, 8\}$ and the draw count as zero.
2. Compute $G(U)$ from the current missing and acquired designs.
3. If $G(U) \leq c$, stop and purchase all designs in U.
4. If $G(U) > c$, pay c and generate one draw.
5. Remove a newly obtained design from U; for an already-owned design, generate its Bernoulli resale outcome.
6. Return to Step 2 until stopping, then record net cost and the realized number of draws.

B Supplementary Diagnostic Figures

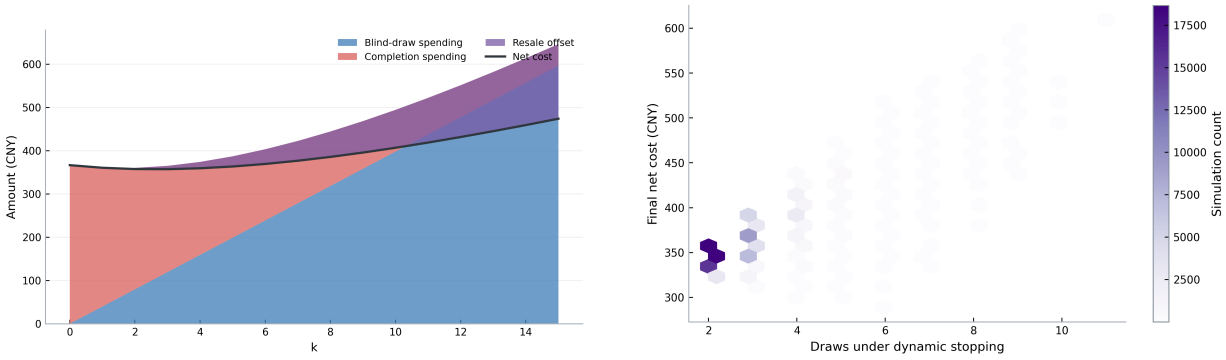
Figures 16 and 17 show draw-count and cost-component diagnostics. Figure 18 compares fixed-strategy distributions and simulation convergence. Figures 19 and 20 show exceedance, sensitivity, and dynamic stopping diagnostics. Figure 21 checks simulation means against theoretical expectations.



(a) Draw count and final cost under collect-all stopping.

(b) Adjacent cost differences for fixed stopping.

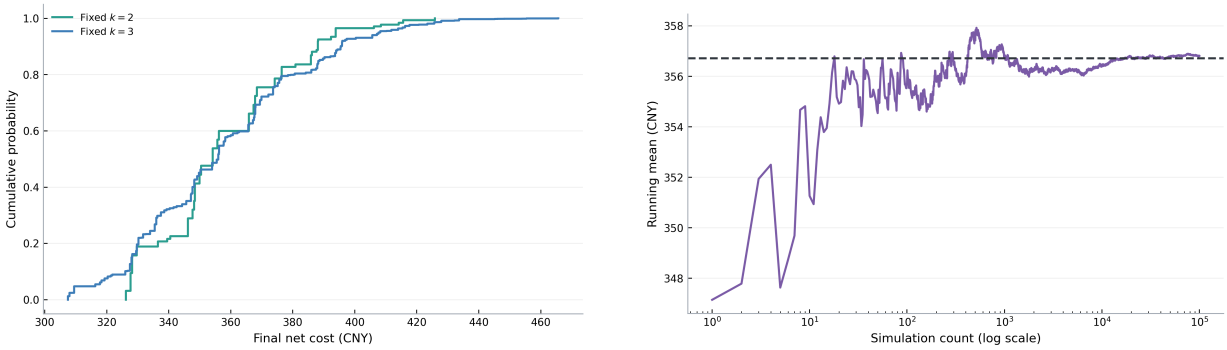
Figure 16. Baseline and fixed-stop diagnostics.



(a) Cost decomposition of the fixed strategy.

(b) Draw count and cost under dynamic stopping.

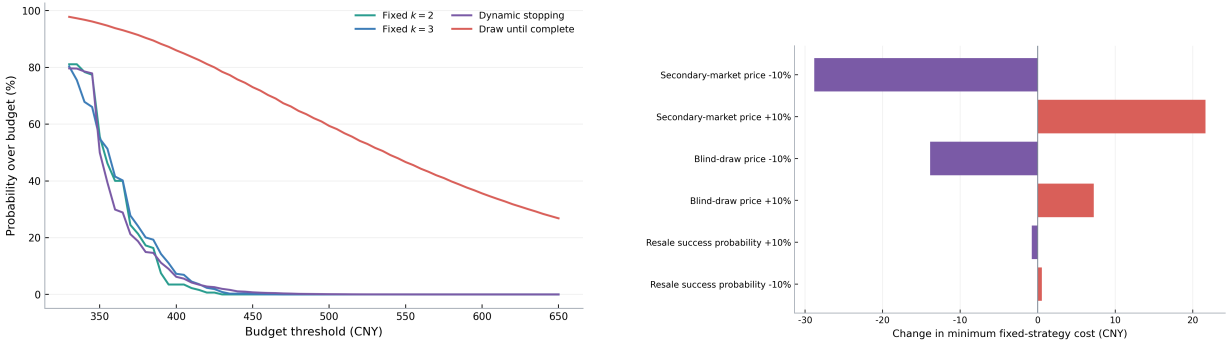
Figure 17. Cost-structure diagnostics.



(a) Empirical CDFs for fixed $k = 2$ and $k = 3$.

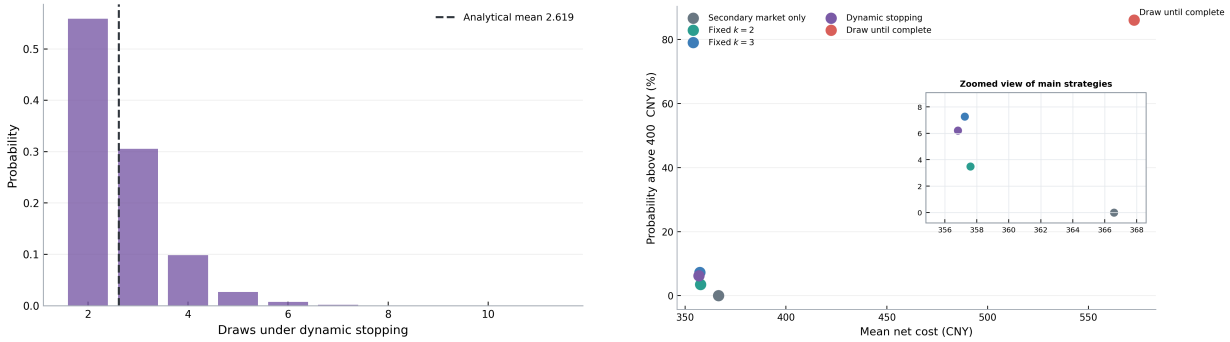
(b) Monte Carlo running mean for dynamic stopping.

Figure 18. Distributional comparison and convergence.



(a) Exceedance probability across budget thresholds. **(b)** Local sensitivity of the optimized fixed cost.

Figure 19. Budget-risk and local-sensitivity diagnostics.



(a) Stopping-time distribution under the dynamic policy. **(b)** Mean cost versus CNY 400 budget risk.

Figure 20. Dynamic stopping and the mean–risk trade-off.

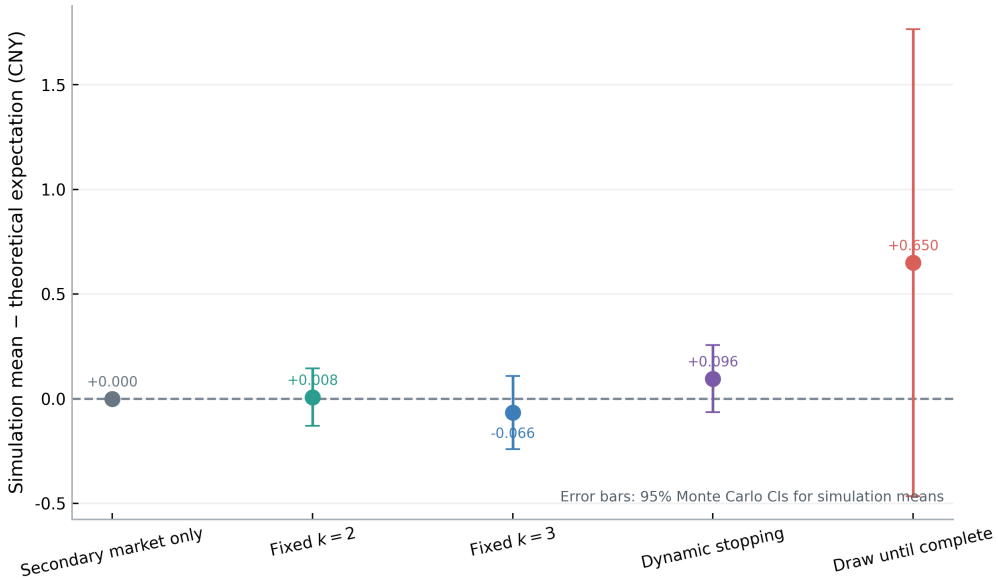


Figure 21. Residuals between Monte Carlo means and theoretical expectations, with 95% confidence intervals.

C All Two-Design Dynamic States

Table 12. Dynamic decisions after two distinct designs have been acquired; monetary values are in CNY.

State	$\delta_i + \delta_j$	$G(U)$	Action	Interpretation
1+2	59.58	38.37	Stop	$G(U) \le c$
1+3	60.39	38.27	Stop	$G(U) \le c$
1+4	59.78	38.35	Stop	$G(U) \le c$
1+5	51.41	39.40	Stop	$G(U) \le c$
1+6	49.09	39.69	Stop	$G(U) \le c$
1+7	44.38	40.28	Continue	$G(U) > c$
1+8	52.32	39.28	Stop	$G(U) \le c$
2+3	59.58	38.37	Stop	$G(U) \le c$
2+4	58.98	38.45	Stop	$G(U) \le c$
2+5	50.60	39.50	Stop	$G(U) \le c$
2+6	48.28	39.79	Stop	$G(U) \le c$
2+7	43.57	40.38	Continue	$G(U) > c$
2+8	51.51	39.38	Stop	$G(U) \le c$
3+4	59.78	38.35	Stop	$G(U) \le c$
3+5	51.41	39.40	Stop	$G(U) \le c$
3+6	49.09	39.69	Stop	$G(U) \le c$
3+7	44.38	40.28	Continue	$G(U) > c$
3+8	52.32	39.28	Stop	$G(U) \le c$
4+5	50.80	39.47	Stop	$G(U) \le c$
4+6	48.48	39.76	Stop	$G(U) \le c$
4+7	43.77	40.35	Continue	$G(U) > c$
4+8	51.71	39.36	Stop	$G(U) \le c$
5+6	40.11	40.81	Continue	$G(U) > c$
5+7	35.40	41.40	Continue	$G(U) > c$
5+8	43.34	40.41	Continue	$G(U) > c$
6+7	33.08	41.69	Continue	$G(U) > c$
6+8	41.02	40.70	Continue	$G(U) > c$
7+8	36.31	41.28	Continue	$G(U) > c$

Design indices follow the order in Table 1. The threshold is $c = 39.8$ CNY.

D Key Results and Reproducibility

Table 13 records key baseline values, including the dynamic results in Eqs. (5.1) and (5.2).

Table 13. Key baseline results.

Quantity	Result
Complete secondary-market purchase	CNY 366.58
Optimal fixed draw count	$k^* = 3$
Fixed-strategy minimum expected cost	CNY 357.3073
Draw-until-complete expected draws	21.7429
Draw-until-complete expected net cost	CNY 571.9538
Dynamic-policy expected draws	2.6190
Dynamic-policy exact expected cost	CNY 356.7156
Bellman initial-state value	CNY 356.7156
Bellman action conflicts	0 of 256 states
Monte Carlo trials	100,000 per randomized strategy

Reproducibility. The separate computational supplement supplies `run_final.py`, the English source workbook, and the prepared price and evidence-mapping files. The entry point reproduces the analysis with base seed 20260917, 100,000 trials per randomized strategy, and 2,000 price-bootstrap resamples. Figures are exported in PNG, SVG, and PDF. Numerical tables use unrounded results, with rounding applied only for display. The supplement README lists the dependencies and run commands.

E Price-Record Traceability

The computational project supplies `market_prices_cleaned.csv`, `market_evidence_mapping.csv`, and `market_trace_audit.json`. The first file contains the prices used in our model. The mapping records the workbook and sheet, row number, source URL, listing cluster, date, and screenshot filename for each observation. The audit file contains the counts and medians from our conservative exclusion check. Row numbers in Table 14 refer to `DEATH_NOTE_My_Dear_Light.xlsx`, worksheet `03_Secondary_Market_Raw_Data`, in the English workbook included in the supplement. Our data-preparation procedure leaves this workbook unchanged.

Table 14 shows three examples of these record mappings.

Table 14. Examples of mappings from calculation records to original price rows.

Evidence ID	Source row	Design index	Recorded date	Unit cost (CNY)
MKT-001	3	1	2026-08-05	45.00
MKT-068	70	3	2026-08-14	61.00
MKT-126	128	8	2025-08-19	68.00

The baseline extraction excludes records 16, 33, 52, 105, 107, 108, and 121 because their inclusion flags reject bundles without a reliable unit price. Records 35 and 98 are also excluded because their unit-price fields are missing despite affirmative inclusion flags. This gives $134 - 7 - 2 = 125$ usable records. We checked every included price, evidence identifier, and date against its workbook row.

The evidence identifiers are distinct, but they are not a guarantee of distinct listings or sellers. The 125 calculation records refer to 114 distinct source URLs. We flag the repeated design and URL in MKT-068/MKT-075 and the unresolved year in MKT-126, retaining their recorded values. Section 6.8 reports the separate 122-record check.

The available fields do not allow us to reconstruct search keywords, result rankings, or seller-level sampling rules. We do not have the original screenshot files in our computational supplement, so the recorded filenames cannot serve as verified screenshot evidence. Our checks establish that the calculation file matches the workbook, but do not establish source-page authenticity, realized sale prices, or random sampling of the market.